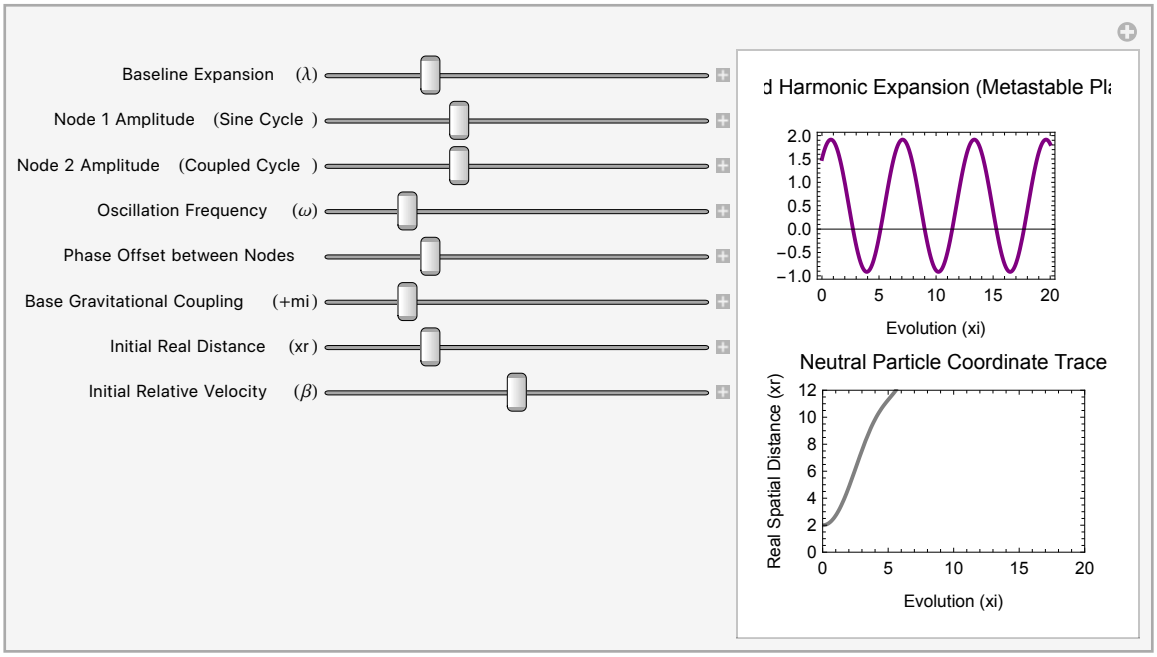
```
In[•]:= Manipulate[
 Module[{tMax = 20, beta, eqOfMotion, solNeutral, metricPressureFunc, x, xi, t},
  (*Normalized velocity*)beta[xVar_, xiVar_] := xVar'[xiVar];
  (*Dual-Node Harmonic Metric Pressure using a distinct localized function name*)
  metricPressureFunc[xiVar_] := baselineExpansion +
    node1Amp * Sin[freq * xiVar] + node2Amp * Sin[freq * xiVar + phaseOffset];
  (*Equation of Motion*)eqOfMotion =
   x''[xi] == metricPressureFunc[xi] - (gForce * (1 + beta[x, xi]^2) / x[xi]^2);
  (*Solver*)solNeutral = Quiet@NDSolveValue[
     {eqOfMotion, x[0] == initialXr, x'[0] == initialBeta}, x, {xi, 0, tMax}];
  (*Plotting the dual-node interference and the resulting geometric trace*)
  (*DO NOT put a semicolon at the end of this Column command*)
  Column[{Plot[metricPressureFunc[xi], {xi, 0, tMax},
     PlotStyle → Directive[Thick, Purple], PlotRange → All, Frame → True,
     FrameLabel → {"Evolution (xi)", "Net Metric Pressure (dp/dxi)"},
     PlotLabel → "Coupled Harmonic Expansion (Metastable Plateaus)"],
    Plot[solNeutral[xi], {xi, 0, tMax}, PlotStyle → Directive[Thick, Gray],
     PlotRange → {{0, tMax}, {0, initialXr + 10}}, Frame → True,
     FrameLabel → {"Evolution (xi)", "Real Spatial Distance (xr)"},
     PlotLabel → "Neutral Particle Coordinate Trace"]}]],
 (*Harmonic Coupling Sliders*){{baselineExpansion, 0.5,
   "Baseline Expansion (λ)"}, 0, 2},
 {{node1Amp, 1.0, "Node 1 Amplitude (Sine Cycle)"}, 0, 3},
 {{node2Amp, 1.0, "Node 2 Amplitude (Coupled Cycle)"}, 0, 3},
 {{freq, 1.0, "Oscillation Frequency (ω)"}, 0.1, 5},
 {{phaseOffset, Pi / 2, "Phase Offset between Nodes"}, 0, 2 Pi},
 (*Kinematic Sliders*)
 {{gForce, 1.0, "Base Gravitational Coupling (+mi)"}, 0.1, 5},
 {{initialXr, 2.0, "Initial Real Distance (xr)"}, 1.0, 5},
 {{initialBeta, 0.0, "Initial Relative Velocity (β)"}, -0.9, 0.9}]
```

Out[ ]=


Baseline Expansion (λ)
Node 1 Amplitude (Sine Cycle)
Node 2 Amplitude (Coupled Cycle)
Oscillation Frequency (ω)
Phase Offset between Nodes
Base Gravitational Coupling (+mi)
Initial Real Distance (xr)
Initial Relative Velocity (β)
d Harmonic Expansion (Metastable Pla
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
0
5
10
15
20
Evolution (xi)
Neutral Particle Coordinate Trace
Real Spatial Distance (xr)
12
10
8
6
4
2
0
0
5
10
15
20
Evolution (xi)